EXP.NO:19

DP - LONGEST NON-DECREASING SUBSEQUENCE

PROBLEM STATEMENT:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

CODE

```c
#include<stdio.h>

int main(){
    int n;
    scanf("%d", &n);

    int arr[n], dp[n];

    for(int i=0;i<n;i++)
        scanf("%d", &arr[i]);

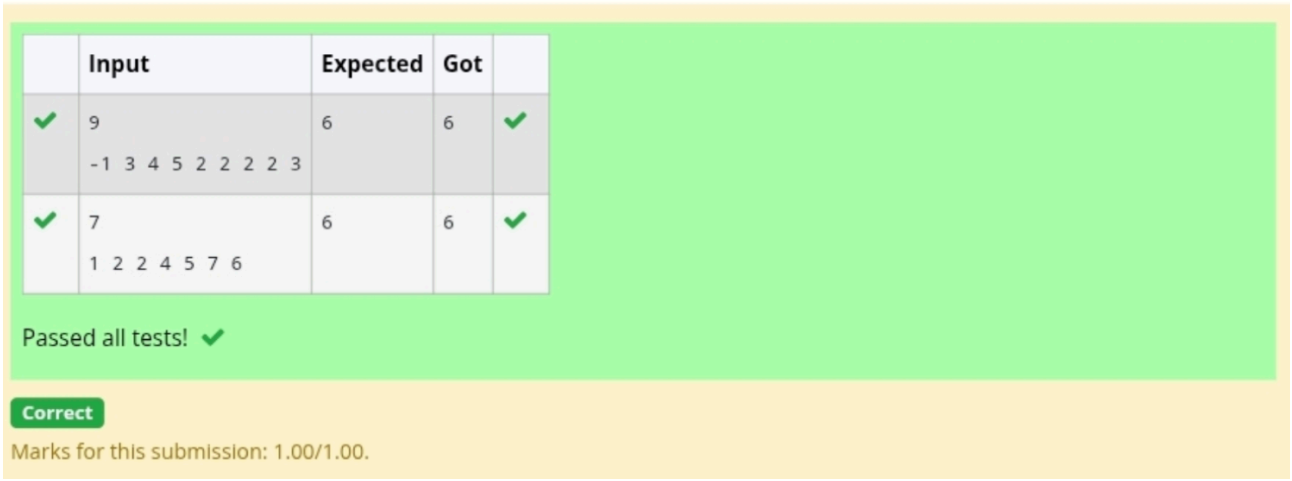
    int max=1;

    for(int i=0;i<n;i++)
        dp[i]=1;

    for(int i=1;i<n;i++){
        for(int j=0;j<i;j++){
            if(arr[j]<=arr[i] && dp[j]+1>dp[i]){
                dp[i]=dp[j]+1;
            }
        }
        if(dp[i]>max)
            max=dp[i];
    }
    printf("%d", max);
    return 0;
}
```

EXP.NO:19

OUTPUT:



	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.